

OpenModelica

OpenModelica is a platform for the Modelica language. It's free, based on open source software and is supported by the Open Source Modelica Consortium. OpenModelica can be installed both on Linux and Windows platforms. In the OpenModelica package, three important modules are available – OMEdit, OMNotebook and OMSHELL. OMEdit is a graphical interface, which can be used to build models and run simulations. OMNotebook is an innovative learning companion, with which you can read the relevant information on a particular concept and also run the code there itself. While it will not teach you how to use OMEdit, which is a graphical editor, OMNotebook will help you to understand the basics of the Modelica language very clearly and help you build a very strong foundation on the topic. You can download and install OpenModelica for Linux from <https://openmodelica.org/download/download-linux>, and for Windows from <https://openmodelica.org/download/download-windows>.

If you face a problem in installing it, you can also download a virtual machine with pre-installed OpenModelica from <https://openmodelica.org/download/virtual-machine>.

Getting started with OMEdit

OMEdit is a graphical editor that allows you to access Modelica's components library, which you can see on the left side of the screenshot in Figure 1. Building a model is as easy as dragging and dropping components into the 'model' area, and connecting suitable components. Of course, you should know which components to use, and what the appropriate value for different parameters of the components is. We'll build a very simple electric circuit to get started with OMEdit. Here are the steps.

1. Start OMEdit.
2. Go to File → New Modelica Class → Name → ResistorCircuit (we've provided model name, not file name).
3. Click on Save → ResistorCircuit.mo [Note: This is the file name].
4. Go to Modelica's components library. Open *Electrical* → *Analog* → *Basic*. Drag and drop resistors into the model area. Double-click, and enter resistor values for both.
5. Similarly, open *Electrical* → *Analog* → *Sources* → *Constant Voltage* (for batteries).
6. Once the components are in place, and the appropriate parameter values are entered, we're ready to connect. To do so, click, drag, and connect from one end of the component to another end.
7. Once all the connections are done, run the simulation.

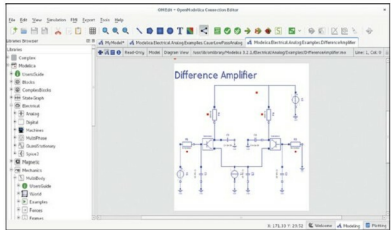


Figure 1: OpenModelica

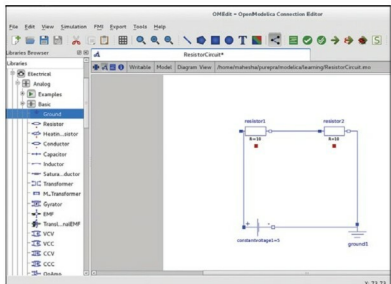


Figure 2: Resistor circuit

- Menu: *Simulation* → *simulate*
8. Viewing the results is also very easy. Just click on the *Plotting* tab in the bottom right corner, and you'll see a list of components in a tree. If you expand the tree, you can select the parameter of your choice, and you'll be able to see the plotting of that parameter value with respect to time. While this example used only electrical components, we can easily use mechanical and electrical components in a single model and simulate the operations. I would also recommend that users explore OMNotebook, as it can be a very good learning companion in the beginning. **END**

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